Team 6 – Software Development Plan

Project Overview

By the end of the semester, we will create a Task Journey: a calendar website that will be tailored towards ADHD and neurodivergent minds with executive dysfunction. This will be done through the use of a gamified to-do calendar, with rewards from task completion used for avatar and calendar customization. Through incentives that will provide external motivation, we can assist ADHD and neurodivergent groups in organizing, managing, and remembering the tasks that they need to complete so that they can be their most successful.

Scope

In-Scope

- RQ-01: The system shall take user input of tasks in the form of text to create events and tasks, which will be added to the user's personal calendar.
- RQ-02: The system shall accept user uploads of downloaded Brightspace calendar information to automatically create events and tasks, which will be added to the user's personal calendar.
- RQ-03: The system shall display events and tasks entered in RQ-01 and RQ-02 through a calendar display and task screen, which can be accessed and viewed by the user.
- RQ-04: The system will allow the user to mark off tasks that are completed, and the user will be awarded Task Tokens, a system currency, based on the difficulty of the task completed.
- RQ-05: The system shall allow users to spend Task Tokens gained from RQ-04 to purchase digital cosmetics and customization items, such as calendar designs.
- RQ-06: The system shall remind the user of upcoming tasks and events through OS notifications and/or emails, with frequency and type set by the user's preference.
- RQ-07: The system shall securely store user tasks, events, and personal information on a cloud database system.
- RQ-08: The system will accept username and password login information from the user, to allow them to access their personal account and calendar.
- RQ-09: The system will allow for the creation of a user account, taking only username and password information.
- RQ-10: The layout of the calendar display and task display mentioned in RQ-02, and other displays such as the total amount of task tokens currently owned, statistics display, and display of completed tasks can be moved according to user preferences, through the use of a widget system.
- RQ-11: The system will have the capacity to play calm music in the background.

- RQ-12: The system shall have user guides and tutorials, to explain different functionalities to the user.
- RQ-13: The system shall allow tasks and events to be edited and deleted.
- RQ-14: The system shall have adequate documentation of use and functionality.
- RQ-15: The system shall run as a web application, accessible through the internet.

Under Evaluation

- Digital cosmetics beyond the calendar customization, including backgrounds, avatars, avatar frames, and more.
    - This will increase user incentives, but due to time constraints, it might not be possible to implement them. If there is enough time for extra development, this will be added to the scope of the system.
- User uploads of Canva and similar downloaded calendar information, beyond Brightspace, to generate tasks and events on the user's calendar.
    - This will be very useful to increase the userbase beyond Brightspace students, however, this may be difficult to implement and test during development. If there is enough time for extra development, this will be added to the scope of the system.
- Pulling info *automatically* from Brightspace, Canva, and similar student calendar applications and websites, to create automatic tasks and events on the user's calendar.
    - The website would be able to automatically register all events, homework assignments, and exam schedules with minimal user input (i.e. a click of a button or a simple connection and sign in feature). There would be major time constraints on how to implement this. A lot more complex and easier said than done to be able to enable a website to automatically pull from a user's personal academic account.
- Local App (with automatic cloud updates when available)
    - The team works on MacOS and Windows computers so it would complicate the process of developing an app. We are choosing to focus on a website-based program for the time being due to this and time constraints. We will consider implementing a local app to the system scope if there is enough time for development.

Out of Scope

- Co-Op Integration
    - Multiplayer integration is out of the scope of our project and unimportant to the overall purpose, as this is not intended to be a social application.
- Minigames
    - Our program should focus solely on the calendar of mechanics and will not contain any additional minigames within the system. This would only serve to distract users from completing the tasks they are trying to complete.
- In-App Purchases
    - Defeats the purpose of our project, as there is no point in completing tasks if the incentives can be purchased. This would likely push students away from wanting to use the site to begin with.
- Focus Features
    - Our program is not meant to force the user to focus. It should not seize control of the user's device in any way, shape, or form. It is only meant to add additional motivation to complete tasks.

Assumptions

- We will work under an honor system: that the user will honestly mark down their tasks as complete and not 'cheat' to get a reward.
- This system's userbase will be on the younger side: high school and college students (15-25).
- This system's userbase has easy access to the internet, as our program is web-based and requires internet access to use.

Constraints

- Language constraints – Our program will only have an English language setting.
- Time constraints – There is a limited amount of time to develop this product.
- Internet requirements – Our application will be limited by network connection as it is an online webpage.
- Budgeting constraints – There is a boundary on how much budget can be spent on APIs, cloud storage space, and other online licensing.

Software Requirements

Functional:

SRID: RQ-01

- ID: FR-01
  - Title: Manual Task Input
  - Statement: This system shall take user input in the form of text and selection, with fields such as due date/date, event/task name, difficulty, description, along with the user's ID, and store them in the AWS DynamoDB calendar event table.
  - Rationale: A manual input is necessary to allow users to add tasks and events.
  - Test Method: Input at least fifteen tasks manually and check that they appear in the database table. Verify that each input is traceable to the respective user.
  - Supporting Context: Assumes each user has a unique ID, and each input item does not exceed the size limit of 400KB.
  - Trace: RQ-101: Event Storing
  - Priority: Must Have

SRID: RQ-02

- ID: FR-05
  - Title: Brightspace Calendar Parsing and Data Entry
  - Statement: The system shall parse the ".ics" file after the user uploads it, and map the fields "DTEND", "SUMMARY", and "DESCRIPTION", to due date, title, and description, which will be used to create a task that is entered into the DynamoDB calendar tasks table.
  - Rationale: Users must be able to recognize their assignment when seeing it in the calendar, with the aforementioned fields.
  - Test Method: Input a Brightspace calendar to the system and check the table to see if it populated the task information correctly.
  - Supporting Context: Brightspace allows users to download an ".ics" file, which is a text file that stores information necessary for calendar events.
  - Trace: RQ-201: Calendar Data Parsing
  - Priority: Must Have

SRID: RQ-02 (cont.)

- ID: FR-06

- Title: Avoid Collisions for Brightspace Calendar Reupload
- Statement: When uploading the parsed .ics event data into the DynamoDB, check the database for an event with the same name for the user. If there exists an event with the same name and date, discard the new event. If there exists an event with the same name but different due date, queue up those events and ask the user whether or not they would like to override tasks whose due dates have changes. If the user responds no, discard the new events; if the user responds yes, change the due date and description for the corresponding events.
- Rationale: Users should be able to import Calendars multiple times and not experience unwanted duplicated tasks. Additionally, due dates may change for assignments, and the old due dates should be changed while keeping the difficulty if the user has already set it.
- Test Method: We will upload the same calendar twice and see if there are duplicates in the data table. We will also create a task with the same name but different due date as a Brightspace assignment, then upload the Brightspace calendar to see if the event's due date got moved.
- Supporting Context: Assume there are no two assignments with the same name in the importing calendar file.
- Trace: RQ-202: Avoid Duplicate Items
- Priority: Medium High

SRID: RQ-03

- ID: FR-04
  - Title: Calendar Event Retrieval
  - Statement: After an event is inputted, deleted, or modified, the calendar waits for DynamoDB's handshake response, and updates the display accordingly, showing events in the when items are successfully inputted and removing the events when items are deleted.
  - Rationale: The calendar must show the events for the correct time and date with the correct text or description to the user by retrieving event data from the DynamoDB table. The calendar display must respond to changes in the database.
  - Test Method: Input fifteen tasks and check that they appear in the calendar. Edit those events and check if they are synced in the calendar. Delete all tasks and check if they are deleted from the calendar.
  - Supporting Context: Assumes DynamoDB responds back after data is inputted or removed. Assumes the user is online, and there are no errors on the database's end.
  - Trace: RQ-301: Calendar Event Sync

- Priority: Must Have

SRID: RQ-04

- ID: FR-07
  - Title: Task Completion
  - Statement: When marking a task as complete from the calendar, the project sends a request to DynamoDB to delete the event in the calendar event table and another request to add the points (corresponding to difficulty) to another DynamoDB table.
  - Rationale: Checking off tasks as complete will need to grant the user a reward and to delete the task from the calendar and data table.
  - Test Method: We will check off 15 tasks of varying difficulty as complete, and check in the data table whether the increase in currency is reflected.
  - Supporting Context: Assume tokens correspond one-to-one with difficulty and do not have to be calculated externally. Assume no errors on DynamoDB's end. Assume data size does not exceed 400KB.
  - Trace: RQ-401: Task Completion Checkoff
  - Priority: Must Have

SRID: RQ-05

- ID: FR-08
  - Title: Token Spending
  - Statement: The system shall allow users to spend Task Tokens gained from RQ-04 to purchase digital cosmetics and customization items, such as calendar designs.
  - Rationale: To motivate a user to succeed, an in-application reward system needs to exist. Having a currency and different prices attached to different customizations will provide additional motivation to gain the points needed to buy each customization they want.
  - Test Method: We will be given 25 Task Tokens which will then be spent on at least one customization item. This requirement will be validated if all purchases of a customization item allows the user to use that item.
  - Supporting Context: Assuming there are no errors with visibility of the website.
  - Trace: RQ-501: Token Spending Verification
  - Priority: Must have

SRID: RQ-06

- ID: FR-09
    - Title: Notification System
    - Statement: The system shall remind the user through OS notifications and emails of any events or tasks that are approaching.
    - Rationale: Since the function of the application is to keep users on track for their due dates, notifications that remind the user how much time they have remained and what rewards they stand to gain by completing the task. This is imperative to motivating users so that they can succeed.
    - Test Method: Have multiple weeks of testing where different users have the same list of due dates that are interspersed throughout. We will check that all notifications were sent at the right time to the right person.
    - Supporting Context: Assuming there are multiple users of the website at the given time when notifications are sent.
    - Trace: RQ-601: Notifications going to the correct user.
    - Priority: Must Have

SRID: RQ-06 (cont.)

- ID: FR-10
    - Title: Frequency of Notifications
    - Statement: The user shall be able to set the frequency at which they receive notifications.
    - Rationale: Users know when in relation to when a task is due they should receive a reminder in order for it to be both effective and motivating.
    - Test Method: During testing we will set our frequency to five days prior, five hours prior, and five minuets prior. Then once it has been validated that those reminders all came at the time they should. Then we will change the frequency for a different set of tasks and verify the notifications came at the correct time.
    - Supporting Context: Assumes the
    - Trace: RQ-602: Notifications going at user chosen times.
    - Priority: Must Have

SRID: RQ-07

- ID: FR-03
    - Title: Task management error handling

- Statement: During user input, deletion, or modification of events, if DynamoDB sends an error, display an error message in the front end and prompt the user to input again. If there is no response from DynamoDB within 1 second, display the error message.
- Rationale: DynamoDB may fail in rare cases, and error management is necessary to ensure no loss of data or crashing.
- Test Method: Send a data request with no internet so that DynamoDB doesn't respond and see if the system displays an error. Send a data request with no parse key to DynamoDB so it sends an error, and see if the error message is displayed on the front end.
- Supporting Context: Assumes a cloud-based database that runs non-locally. DynamoDB is generally reliable but may fail in rare cases.
- Trace: RQ-701: Database Error Management
- Priority: Medium

SRID: RQ-09

- ID: FR-13
  - Title: User Login
  - Statement: The system will allow for the creation of a user account, taking only username and password information.
  - Rationale: To keep user information separate and secure, a login is required.
  - Test Method: We will make sure that there is no way to access any page beyond the login page. We will also try to make logins with duplicate usernames which should not be allowed to be made.
  - Supporting Context: Assume usernames are user emails.
  - Trace: RQ-901: Login
  - Priority: Must Have

SRID: RQ-09 (cont.)

- ID: FR-14
  - Title: Password complexity
  - Statement: Passwords must be at least 8 characters and have at least one special character.
  - Rationale: A complex password is necessary to keep user information secure from outside attacks.
  - Test Method: We will try to make passwords that do not meet the requirements and check that they are not made into user accounts.

    - Supporting Context: N/A
    - Trace: RQ-901: Login Security
    - Priority: Must have

SRID: RQ-10

- ID: FR-15
    - Title: Display Customization
    - Statement: The layout of the calendar display and task display mentioned in RQ-02, and other displays such as the total amount of task tokens currently owned, statistics display, and display of completed tasks can be moved according to user preferences, through the use of a widget system.
    - Rationale: For the user to be their most successful, the layout of the site should show what is the most intuitive to the user. The user can then customize the layout to their liking.
    - Test Method: We will move all moveable items to all possible locations on the page. We will then check that all data is appearing as it should no matter the location.
    - Supporting Context: Assumes the user wants to customize beyond the original template provided.
    - Trace: RQ-1001: Website Customization
    - Priority: Must Have

SRID: RQ-11

- ID: FR-12
    - Title: Embed Music File in HTML
    - Statement: The project shall have audio file(s) in MP3 or WAV form embedded using the <audio> tag to the webpage and loops.
    - Rationale: The audio files are stored locally and embedded in the page for simple playback.
    - Test Method: Open the project web page and see if there is sound playing. Check that the audio loops after it finished playing.
    - Supporting Context: Assume audio files are short, under 5 minutes. We will make at least one piece of original music, between 1-5 minutes, looping.
    - Trace: RQ-1101: Music Playback
    - Priority: Low Medium

SRID: RQ-12

- ID: FR-11
    - Title: Display Information for Each Element
    - Statement: When the user's mouse hovers on an element such as a button, statistic, or layout, display text at the bottom of the screen with a brief description of the element's function. The user can enable this mode by toggling an information button.
    - Rationale: Users may want information on the function of the calendar even outside tutorial mode.
    - Test Method: Toggle on information mode and hover the mouse over various elements like the token display, calendar view, to-do list view to see if the text shows up.
    - Supporting Context: Assuming we will write descriptions for each functional element in the calendar.
    - Trace: RQ-1201: Information Display
    - Priority: Low Medium

SRID: RQ-13

- ID: FR-02
    - Title: Task Modification
    - Statement: The system shall allow users to edit events. During editing, users may change any text box in the event description, title, and time. After saving, the system sends the request to DynamoDB where the event data is updated in the table.
    - Rationale: Users must be able to easily edit their events without having to delete and re-input events.
    - Supporting Context: There will be an option to edit events when clicking on one in the system. The system will display the input view but with all fields already populated with the previous input.
    - Trace: RQ-1301: Task Editing
    - Priority: Must Have

Non-functional:

SRID: RQ-02

- ID: NFR-04
    - Title: Usability of External Calendar Integration
    - Statement: The system shall explain how external calendars can be imported into the website.
    - Rationale: Users may not be familiar with how to download static versions of other calendar applications. The website will provide them with this so that they do not have to waste time manually importing tasks because they do not know how to import other calendars.
    - Test Method: We will ensure that first time users can both find and understand the instructions on how to
    - Supporting Context: Assume that they are importing from Brightspace or Google calendar.
    - Trace: RQ-203: Calendar Importability
    - Priority: Must have

SRID: RQ-07

- ID: NFR-03
    - Title: Database security and accessibility
    - Statement: The system shall securely store user tasks and personal information on a cloud database system through AWS DynamoDB, which can be viewed by the user from within the application. Through AWS DynamoDB, the system will implement server-side encryption, and Transport Layer Security (TLS) for API calls.
    - Rationale: Because all data is being stored remotely it is necessary to maintain confidentiality and accessibility to only the right people to follow both ethical and legal requirements.
    - Test Method: We will attempt to access personal information and task information when not logged in through a variety of methods.
    - Supporting Context: Assume AWS DynamoDB is secure.
    - Trace: RQ-701: Database Security Checks
    - Priority: Must Have

SRID: RQ-08

- ID: NFR-02

    - Title: Website Security
    - Statement: The system will accept username and password login information from the user, to allow them to access their personal account and calendar.
    - Rationale: Ensuring security of user information is vital to follow ethical and legal
    - Test Method: At least three attempts of incorrect login will be made on test credentials. A password cracking program will also run over the login. The site will be checked for any potential vulnerabilities.
    - Supporting Context: Assumes the required password is necessarily complex.
    - Trace: RQ-801: Website Login Requirements
    - Priority: Must Have

SRID: RQ-14

- ID: NFR-05
    - Title: Website Maintenance
    - Statement: The website shall be maintained through user and developer documentation.
    - Rationale: User and developer documentation is vital for a system to be maintainable. Documentation ensures that the end goal is being upheld even if the development team changes and limits the amount of work that is needed for the new team to figure out how the application works.
    - Test Method: Having non-development team members read through the documentation to ensure that it makes sense and could be used if there were a change in development team.
    - Supporting Context: Assumes that those reading the developer documentation have a background in software.
    - Trace: RQ-1401: Web Page Maintenance
    - Priority: Must have

SRID: RQ-15

- ID: NFR-01
    - Title: Web page reachability
    - Statement: This system must be reachable through the web application at all times.
    - Rationale: If the site is not reachable, the user will not be able to use any functionality of the application.
    - Test Method: Have a minimum of twenty IP's access the site at once to ensure that the site will not crash due to an uptick of web traffic.
    - Supporting Context: Assumes that the user has internet connection.

- Trace: RQ-1501: Web Page Accessibility
- Priority: Must Have

Deliverable:

Deliverable: A fully deployed, publicly accessible calendar web application with user login support and all core project features implemented.

Requirements: NFR-01, NFR-04, FR-13, RQ-15

Deliverable: A configured AWS DynamoDB database for storing and retrieving calendar event data. The deliverable will include the database table configuration, database access code, and documentation describing the table structure, data fields, and procedures for creating, retrieving, and managing calendar event records.

Requirements: NFR-03, FR-04

Deliverable: In-website user documentation explaining how the different functions of the applications work, and an in-codebase developer documentation such that the application can be maintained even if there are changes to the development team.

Requirements: NFR-05, RQ-14

Standards:

ISO 27001: This standard is applicable to this system because we will be storing user information. Following a standard on cyber safety is important to the trust of users. This will impact our design and code by ensuring that we have additional security measures on all aspects of our code to ensure that user information is kept secure. We plan to ensure compliance by using Transport Layer Security as well as requiring users to make complex passwords. We will also make sure all software updates for services are up to date.

ISO 8601: This standard is applicable to our project because we use dates and times in the tracking of tasks. It is important for our design to have a standardized date format so that reminders are all sent out at time. It will change how we make our design as will have to ensure that all inputs of date and time are in the correct format. We will ensure that this standard is followed by preforming an initial data transform on all dates and times coming into the application and also preforming multiple checks throughout the pipeline.

ISO/IEC 30071: This standard is applicable to our project because we are building this website to be accessible to everyone. This will change our design process as following this standard will

change priorities to ensure accessibility. We will make sure there are proper accessibility options such as alternative fonts and text size. We will also make sure to document accessibility choices in both user and developer documentation.

GenAI Statement

GenAI was used to draft the second deliverable, in addition to spell-check for the documentation.

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